

HAO LIU

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⚙️ QUALIFICATIONS SUMMARY

- Tech lead with 9+ years designing and shipping AI systems across computer vision, machine learning, and large language models, with focus on improving engineering productivity and troubleshooting efficiency.
- Demonstrated innovative thinking in turning vague problem spaces into end-to-end solutions: from knowledge-base / graph design and retrieval to multi-stage reasoning, evaluation, and continuous improvement loops.
- 9+ years of hands-on development in Python and C/C++ with solid Object-Oriented Programming (OOP), comfortable moving between low-level implementation details and high-level system/architecture thinking.
- Extensive experience with generative models (GANs, diffusion models, LLMs, VLMs) and LLM tooling, including PyTorch, DeepSpeed/FSDP, vLLM/SGLang, scikit-learn, OpenCV, and deep learning compilers.
- Proficient with AWS, GCP, and Azure, and with Kubernetes and Docker for deploying and scaling production ML/LLM and developer-productivity services.
- Track record of leading cross-functional projects from concept to production, aligning stakeholders, mentoring engineers, and driving adoption of new AI capabilities in mission-critical workflows.
- Innovator with strong publication and patent contributions and a consistent history of turning research ideas into robust, revenue-impacting industrial solutions.

👤 EXPERIENCE

Senior Software Engineer, Computer Vision and Machine Learning at **KLA Corporation**

Jul. 2016 – Present

1. KLA SemiCon-Assistant System

– Generative AI & LLM System Architecture

- **Enterprise GenAI Troubleshooting Platform:** Architected and led launch of **SemiCon-Assistant**, an enterprise-scale GenAI troubleshooting platform that ingests a multi-decade corpus of unstructured/semi-structured service data to generate actionable, context-aware resolutions, transforming multi-week semiconductor equipment support cycles into near real-time, self-service workflows for field engineers.
- **Retrieval-Augmented Generation (RAG & GraphRAG):** Designed end-to-end RAG stacks over the same 150K+ documents and 40K+ tickets, evolving from naive dense/sparse RAG (LlamaIndex + Elasticsearch vector kNN/BM25) to graph-based RAG using Neo4j, Microsoft GraphRAG, and LightRAG, enabling structured reasoning over component, subsystem for more precise troubleshooting.
- **Multi-Stage Intelligence Pipelines:** Built production multi-stage retrieval and reasoning pipelines combining generative query expansion/rewrite, step-back prompting for higher-level abstraction and intent understanding, and a two-phase recall+reranker architecture: high-recall parallel dense (vector) / BM25 hybrid search with metadata-aware filtering/boosting, followed by KG-aware subgraph assembly, Cross-Encoder reranker cascades, and deduplication/fusion. This architecture cut hallucinations by 38%, lifted top-1 exact-match accuracy by 32%, and maintained p95 latency under 400ms in production.

– Data Engineering & Processing for AI

- **LLM Data Augmentation & Prompt Optimization:** Built LLM-driven augmentation pipelines that turn proprietary tickets/logs into high-quality instruction datasets using few-shot and chain-of-thought prompting with GPT-4 and Claude, and applied DSPy/TextGrad to automatically tune prompts, few-shot examples, and RAG/agent module configurations from dev-set metrics and textual feedback, reducing manual prompt work and improving task accuracy and robustness.
- **Human-in-the-Loop (HITL):** Designing and managing HITL workflows with domain experts to validate, clean, and score synthetic data, ensuring high relevance and quality for model fine-tuning.

– LLM Fine-Tuning, Alignment & Safety

- **Fine-Tuning Strategies:** Applying full fine-tuning (FT), and parameter-efficient fine-tuning (PEFT) techniques (e.g., LoRA, QLoRA, adapter-based methods) to adapt base models to domain-specific tasks while controlling compute and deployment costs.

- **Reinforcement Learning & Continuous Improvement:** Building closed-loop systems for continuous model improvement using Reinforcement Learning from Human Feedback (*RLHF*) and Reinforcement Learning from AI Feedback (*RLAIF*), systematically converting human and AI critiques, failure cases, and preference data into new fine-tuning or PEFT cycles.
 - **AI Safety & Red Teaming:** Developing and deploying automated red-teaming modules to test for and mitigate vulnerabilities such as prompt injection, jailbreaking, data leakage, and model biases.
- **LLM Evaluation & Deployment**
- **Comprehensive Evaluation Frameworks:** Establishing robust evaluation systems combining offline metrics (e.g., *RAGAS* for faithfulness, context precision) with live online A/B testing infrastructure.
 - **Performance Monitoring:** Establishing and tracking performance indicators such as top-1 exact-match accuracy, factual consistency, and end-to-end latency (e.g., maintaining *p*95 latency under 400ms).
 - **Auto-scaling & Deployment:** Using Kubernetes and Amazon Elastic Kubernetes Service (Amazon EKS) to deploy LLM services with horizontal auto-scaling, rolling updates, and health-based restarts, ensuring reliable and cost-efficient operation under variable load.
- **High-Performance LLM Infrastructure & Optimization**
- **Inference Acceleration:** Engineered high-throughput, low-latency serving stacks for billion-parameter Transformers, delivering up to 7× throughput via full-stack optimization: disaggregated prefill/decode with KV caching, PagedAttention, I/O-aware fused kernels (FlashAttention), continuous batching, architectural choices (MQA/GQA/MLA), and model compression (quantization, pruning, distillation).
 - **Large-Scale Distributed Training:** Accelerated large-scale LLM training using a full suite of parallelism strategies (DP, TP, PP, SP, EP) and DeepSpeed ZeRO for memory optimization, combined with mixed precision (BF16/FP16) to maximize hardware utilization and reduce end-to-end training time.

2. RAPID SemiConductor Wafer Inspection Equipment Project

- Semiconductor image data is hard to get due to customer's IP policy and expensive to craft (very different characteristic with natural image). Built a semiconductor specific image augmentation pipeline which integrated GAN, latent space GAN, and diffusion models to generate high resolution semiconductor wafer images with realistic textures whose micron scale structures are missing from natural image data. By injecting edge and depth cues parsed from design schematics into conditional GAN/diffusion generators, reduced manual data collection effort by 70% and delivered markedly higher fidelity images that drive superior defect detection and image classification accuracy in large backbone deep learning models.
- Defect Localization – Stage 1: Designed a sub-micron wafer-defect detector that ensembles DETR, YOLOv8, and SSD. Custom anchor tuning for YOLOv8 and SSD, combined with focal-loss re-weighting, pushed mean localization recall to 94%, enabling bounding boxes even on sparse, low-contrast patterns.
- Defect Classification – Stage 2: Fine-tuning ResNet-50, DenseNet-121, EfficientNet-B3, and ViT-B/16 on the localized crops, leveraging mixup, CutMix, and semiconductor specific augmentations to build an ensemble classifier with a 0.92 macro F1 across 128 defect categories.
- The two stage wafer defect analysis framework eclipsed the incumbent vision stack in 107 / 128 classes and lifted overall inspection accuracy by +2.3 pp, directly improving yield loss forecasting and accelerating root cause analysis for high volume semiconductor fabs.

3. Zeta3D Optical Profilometer Project

- Principal developer of the Zeta3D Phase Shifting Interferometry (PSI) scanning mode. Resolved a long-standing phase unwrapping issue by implementing a novel 3D sigma filter, a bad-pixel-filling algorithm to handle noisy and null data, and a 2D feature detection algorithm. These contributions dramatically improved the metrology system's accuracy and robustness specifications.
- Proposed a novel, robust, and fast two-dimensional phase unwrapping method (RF2D-SRNCP) to overcome traditional phase unwrapping issues for KLA's ultra-high accuracy Zeta3D optical metrology profiler.
- As a leading contributor, summarized the work in a paper titled "A Robust and Fast Phase Unwrapping Algorithm for Phase Shift Interferometry Scanning Mode." Presented this research at the 2018 KLA Engineering Conference in Monterey, California.
- This work was successfully integrated into the Zeta3D optical profiler tool, contributing to approximately \$10M in annual revenue since 2019. Received a team award for the successful Zeta-388 project launch.

4. MicroXAM-800 Optical Surface Profiler Project

- Roughness and waviness analysis is critical to surface metrology for semiconductor industry wafer inspection. Methodologies follow standards like [ISO 4287:1997](#), [BS EN ISO 4287:2000](#) and [ISO 21920-2:2021](#). Our work overcame significant processing bottlenecks from proprietary image filters for detecting fine-grained wafer defects, enabling real-time computation and review for large-scale wafer image datasets.
- Traditionally, calculating roughness/waviness using direct spatial convolution with large filters (e.g., 251×251 or 511×511) is computationally expensive. For a standard 1024×1024 image, this process can take several minutes on a CPU with an unoptimized algorithm.
- The Convolution Theorem dictates a clear trade-off based on filter size. While spatial domain convolution is more efficient for small kernels with a complexity of $O(N^2M^2)$, frequency domain convolution using the Fast Fourier Transform (FFT) is superior for large kernels, with a complexity of just $O(N^2 \log N)$. This principle identifies the computational crossover point that guided our optimization.
- Algorithm optimized for CPU/GPU deployment: CPU implementation with 2D-FFTW reduces 1024×1024 image processing to ~650 ms, while a CUDA-based GPU version handles 10k×10k images in ~2s.

5. Profiler P7/P17/P170/HRP-260 Project

- Enhanced quality and uniformity of live video from KLA's laser-based CCD optical systems, mitigating effects of malfunctioning automatic illumination controls and ensuring a consistent, high-quality feed that supports robust downstream computer vision and machine learning performance.
- Developed computer vision algorithms to identify Critical Dimensions (CD) within a Region of Interest (ROI), compensating for positioning errors from mechanical stage drift and guiding KLA's high-resolution automated stylus to the exact target. This enables precise nanoscale measurements of semiconductor features, vital for process control in engineering and manufacturing.
- Augmented the existing AutoFocus system for KLA's optical setup by implementing advanced image processing techniques. The system now uses saliency detection to simplify the image and isolate the most prominent objects for focusing, which is combined with automatic boundary detection and signal processing algorithms that sharpen object edges to produce a consistently crisp and accurate focus.

PUBLICATIONS

- Hongjun Su and **Hao Liu**. “[A novel dynamic classifier selection algorithm using spatial-spectral information for hyperspectral classification](#).” *Remote Sensing for Natural Resources* 29, no. 2 (2017): 15-21.
- Su, H.; Yong, B.; Du, P.; **Liu, H.**; Chen, C.; Liu, K. “[Dynamic classifier selection using spectral-spatial information for hyperspectral image classification](#),” *J. Appl. Remote Sens.* **8**(1):085095, 2014.
- **Hao Liu**, Dongyang Zhang and Jian Feng. “[Error Analysis for Data of Terrestrial 3D Laser Scanner](#).” *Journal of Water Resources and Architectural Engineering*. 2012, 10(04): 38-41.
- **Hao Liu**, Dong Li, Zhenguo Yu and Xu Jiang, “[Boundary Point Extraction of Point Cloud Based on Range Image](#).” *Journal of Water Resources and Architectural Engineering*. 2013,11(03):156-158.
- **Liu, H.**; Xu, N. “[Exploration and Study for GPS Elevation Abnormalism Fitting Model Based on Wavelet Neural Network](#).” *Journal of Water Resources and Architectural Engineering*. 2013,11(02):182-184+203.
- **Hao Liu**, Bin Luo and Liangpei Zhang. “[Image Simplification based on Saliency Detection](#).” *Journal of Yunan Minzu University (Natural Science Edition)* 25, No. 3 (2016): 257-263.
- Patents holder: [CN202434062U](#), [CN202523220U](#), [CN202523214U](#), [CN201883919U](#), [CN202520595U](#)
- Invited reviewer for a list of reputable journals, conferences and books:
 - [Linear Algebra and Optimization for Machine Learning: A Textbook \(2020\)](#)
 - [IEEE Transactions on Neural Networks and Learning Systems](#)
 - [IEEE International Conference on Image Processing](#): ICIP'23, ICIP'24 TOP 10 Reviewer, ICIP'25
 - [The International Symposium on Visual Computing](#): ISVC'23, ISVC'24, ISVC'25
 - [The British Machine Vision Conference](#): BMVC'23, BMVC'24 OUTSTANDING Reviewer, BMVC'25

EDUCATION

Wuhan University, Wuhan, China

Sept. 2013 – Jun. 2016

M.S. in Geographic Information System (GIS), Photogrammetry and Remote Sensing

Hohai University, Nanjing, China

Sept. 2009 – Jun. 2013

B.S. in Geographic Information System (GIS), Geodesy and Geomatics Engineering